AWS Systems Manager Session Manager: Replacing SSH

Secure, Centralized, and Auditable Access to AWS EC2 Instances

Project Report

Environment: AWS Learner Lab

Region: us-east-1

Prepared for Academic Project Demonstration

1. Abstract

This project demonstrates the use of AWS Systems Manager Session Manager as an alternative to traditional SSH-based access for Amazon EC2 instances. Traditional SSH commonly requires inbound TCP port 22 and relies on SSH key-pair management. The implemented solution uses Session Manager and the SSM Agent to provide interactive access to the same EC2 instance without requiring inbound SSH access. Session activity was configured for centralized logging using Amazon CloudWatch Logs and Amazon S3.

The implementation was validated in stages: traditional SSH access was first tested with port 22 enabled; Session Manager access was then configured and verified; CloudWatch and S3 session logging were enabled; and finally the inbound SSH rule was removed from the EC2 Security Group while Session Manager access remained available.

2. Introduction

Amazon EC2 provides virtual servers that organizations commonly manage remotely. SSH is a standard method for remote administration, but as the number of servers and users increases, organizations may face additional effort around SSH key management, inbound port exposure, centralized access control, auditing, and bastion-host administration.

AWS Systems Manager Session Manager provides an AWS-integrated method for interactive instance access. In this project, Session Manager is used to demonstrate access to an EC2 instance without relying on inbound SSH port 22, while session information is stored for monitoring and auditing.

3. Problem Statement

Traditional SSH access can create operational and security-management challenges when used across many servers and developers. These include the need to manage SSH keys, permit inbound SSH traffic, control access consistently, and maintain centralized evidence of remote sessions.

The project therefore investigates a practical alternative using AWS Systems Manager Session Manager.

4. Objectives

- Create and configure an Amazon EC2 instance for remote administration.
- Demonstrate traditional SSH access through TCP port 22.
- Configure and verify AWS Systems Manager Session Manager.
- Verify the SSM Agent running on the EC2 instance.
- Enable session logging to Amazon CloudWatch Logs.
- Enable session logging to an Amazon S3 bucket.
- Remove inbound SSH/TCP port 22 and verify that Session Manager access continues to work.
- Compare the operational and security characteristics of traditional SSH and Session Manager.

5. AWS Services and Technologies Used

Service / Technology	Purpose in the Project
Amazon EC2	Provides the virtual server being remotely managed.
AWS Systems Manager	Provides management capabilities for the EC2 instance.
Session Manager	Provides interactive shell access without requiring inbound SSH access.
SSM Agent	Runs on the EC2 instance and communicates with Systems Manager.
AWS IAM	Provides identity and authorization mechanisms for AWS access.
Security Groups	Control inbound and outbound network traffic for the EC2 instance.
Amazon CloudWatch Logs	Stores and monitors Session Manager session logs.
Amazon S3	Provides persistent object storage for Session Manager session logs.
SSH	Used as the traditional baseline for comparison and testing.

6. System Architecture

6.1 Traditional SSH

```
Developer / Mac
|
| SSH over TCP port 22
v
EC2 Security Group
|
v
Amazon EC2
```

The traditional baseline used an SSH key pair and an inbound Security Group rule for TCP port 22. The existing EC2 instance was successfully accessed through SSH before changing the inbound rule.

6.2 Proposed Session Manager Architecture

```
Developer
|
v
AWS Console / CLI
|
v
AWS Systems Manager
|
v
Session Manager
|
v
SSM Agent
|
v
Amazon EC2
|
+----------+----------+
| |
v v
CloudWatch Logs S3
```

The final demonstration removed inbound TCP port 22 from the Security Group. Session Manager continued to provide interactive access through the SSM Agent.

7. Implementation

7.1 EC2 Instance

An EC2 instance named SessionManager-Demo was configured in us-east-1 using Amazon Linux 2023 and a t3.micro instance type. The AWS Learner Lab instance profile, LabInstanceProfile, was attached to the instance.

7.2 Traditional SSH Baseline

Before replacing SSH, the instance was tested using traditional SSH. The Security Group allowed SSH/TCP port 22 from the configured source, and the EC2 public IPv4 address was used for the connection.

```
ssh -i ~/.ssh/labsuser.pem ec2-user@
```

The connection was verified using commands such as whoami and hostname.

7.3 Session Manager

The instance was selected from Systems Manager Session Manager and a browser-based session was started. The session was verified with:

```
whoami
hostname
cat /etc/os-release
```

The Session Manager shell reported the session user as ssm-user.

7.4 SSM Agent Verification

```
sudo systemctl status amazon-ssm-agent
```

The service was verified as active and running.

7.5 CloudWatch Logging

Session Manager preferences were configured to stream session logs to the CloudWatch log group /aws/ssm/session-manager. A log stream was observed containing session information and session activity.

7.6 S3 Logging

An S3 bucket was configured for Session Manager session-log storage. The bucket used encryption and public access remained blocked. A generated log object was observed in the bucket after a Session Manager session was completed.

7.7 Removing Inbound SSH

The inbound SSH/TCP 22 rule was removed from the EC2 Security Group. The final inbound-rules view showed no SSH rule. A new Session Manager session was then used to verify that the same EC2 instance remained accessible.

8. Testing and Results

Test	Expected / Observed Result
EC2 instance status	Running; status checks passed.
Traditional SSH	Successful while inbound TCP 22 was allowed.
Session Manager	Successful interactive session to the same EC2 instance.
whoami in Session Manager	Reported ssm-user.
SSM Agent	Active and running.
CloudWatch logging	Session log group and stream available.
S3 logging	Session log object created in the configured bucket.
Inbound TCP 22 removed	No inbound SSH rule remained.
Session Manager after port 22 removal	EC2 remained accessible through Session Manager.

9. SSH vs Session Manager

Feature	Traditional SSH	Session Manager
Inbound access	Typically requires SSH/TCP 22	No inbound SSH port required
Authentication / authorization	SSH key-based access	AWS IAM-based authorization
Session logging	Requires additional setup	Session Manager logging can be configured
Bastion host	May be used	Can be avoided for this access pattern
Centralized management	More operational effort	Integrated with AWS Systems Manager
Project demonstration	Baseline	Final solution

10. Security Considerations

- Inbound SSH/TCP port 22 was removed after the traditional SSH baseline was demonstrated.
- Security Groups were used to control network access to the EC2 instance.
- IAM and the instance profile provide AWS authorization for Systems Manager functionality.
- S3 public access remains blocked and encryption was enabled for the session-log bucket.
- AWS credentials, private keys, passwords, access keys, and session tokens must not be stored in source control.
- Session logs should be treated as potentially sensitive because they may contain session identifiers or user-related information.

11. Advantages

- Reduces the need for inbound SSH exposure.
- Reduces dependence on manually distributed SSH keys for Session Manager access.
- Provides centralized AWS-based access control.
- Supports centralized session logging through CloudWatch Logs and S3.
- Can reduce operational dependence on bastion hosts for interactive instance administration.
- Provides a scalable management approach for multiple managed instances.

12. Limitations and Considerations

- Session Manager requires a properly configured managed instance and working SSM Agent connectivity.
- AWS IAM permissions must be configured correctly.
- Logging services such as CloudWatch and S3 can have their own usage and storage costs outside a lab environment.
- This project demonstrates the access pattern on an EC2 instance; production environments require broader IAM, network, logging-retention, and governance design.

13. Future Enhancements

- Use VPC endpoints for private Systems Manager connectivity in a private-subnet architecture.
- Define more granular IAM permissions using least privilege and resource/tag-based controls.
- Integrate centralized monitoring and alerting for session activity.
- Define log retention and lifecycle policies for CloudWatch and S3.
- Extend the pilot to multiple EC2 instances and multiple developer roles.

14. Conclusion

The project successfully demonstrated AWS Systems Manager Session Manager as an alternative to traditional SSH access for an EC2 instance. Traditional SSH access was first verified through inbound TCP port 22. Session Manager was then configured and verified using the SSM Agent, and session logging was enabled for CloudWatch Logs and Amazon S3.

The final test removed the inbound SSH rule while a new Session Manager session continued to provide access to the same EC2 instance. This demonstrates the central project objective: interactive EC2 administration can be performed without exposing inbound SSH port 22, while AWS-native authorization and session logging can be used to improve centralized management and auditing.

15. References

- AWS Systems Manager Session Manager — project presentation and implementation material.
- AWS Learner Lab environment documentation provided for the project.
- AWS service documentation for Amazon EC2, AWS Systems Manager, IAM, Amazon CloudWatch, and Amazon S3.